

Semester	B.E. Semester VII – Computer Engineering
Subject	R programming
Subject In-charge	Prof. SDW

Student Name	Diksha Parulekar.
Roll Number	23102A0068.
Grade and Subject Teacher's Signature	

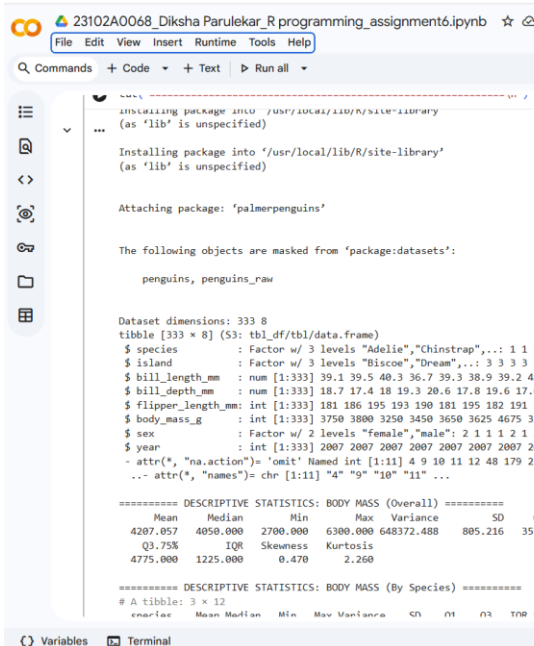
Experiment Number	06	
Experiment Title	Statistical Analysis of Physical Characteristics of Palmer Penguins	
Resources / Apparatus Required	Hardware: Computer system	Software: Python
Description	Github link : dikshaparulekar/23102A0068_R-PROGRAMMING This experiment presents a comprehensive statistical analysis of the physical characteristics of penguins using the Palmer Penguins dataset, sourced through the palmerpenguins R package and originally collected from the Palmer Station Long-Term Ecological Research (LTER) program in Antarctica. After removing rows with missing values, the working dataset contains 333 complete observations across three species — Adélie, Chinstrap, and Gentoo. The key variables used in the analysis are body_mass_g, flipper_length_mm, species, and sex. The analysis began with descriptive statistics for body mass, including mean, median, minimum, maximum, variance, standard deviation, quartiles, IQR, skewness, and kurtosis. These statistics were calculated overall and for each species and visualized using a histogram, species-wise and sex-wise boxplots, and a density plot. Hypothesis testing was performed to determine whether male and female penguins differ significantly in body mass. Normality was assessed using the Shapiro-Wilk test and QQ-plots, followed by an independent two-sample t-test, a 95% confidence interval for the difference in means, and Cohen's d to measure the effect size. A one-way ANOVA was conducted to test whether mean body mass differs among the three penguin species. Normality and homogeneity of variance were checked	

using **Shapiro-Wilk, QQ-plots, and Levene's/Brown-Forsythe test**. Since the ANOVA result was significant, **Tukey's HSD post-hoc test** was used to identify the species pairs with significant differences. A **Kruskal-Wallis test** was also performed to validate the findings without relying on normality assumptions.

A **two-way ANOVA** was then performed to study the individual and combined effects of **species and sex** on body mass, including their interaction effect. An interaction plot was used for visualization.

Finally, the same statistical analysis was repeated for **flipper length** to examine whether the species-level differences found in body mass were also reflected in flipper length. A final group comparison boxplot with an annotated ANOVA p-value was created to summarize the significant differences in body mass among the species.

Program
& Output



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installing package into /usr/local/lib/R/site-library
(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

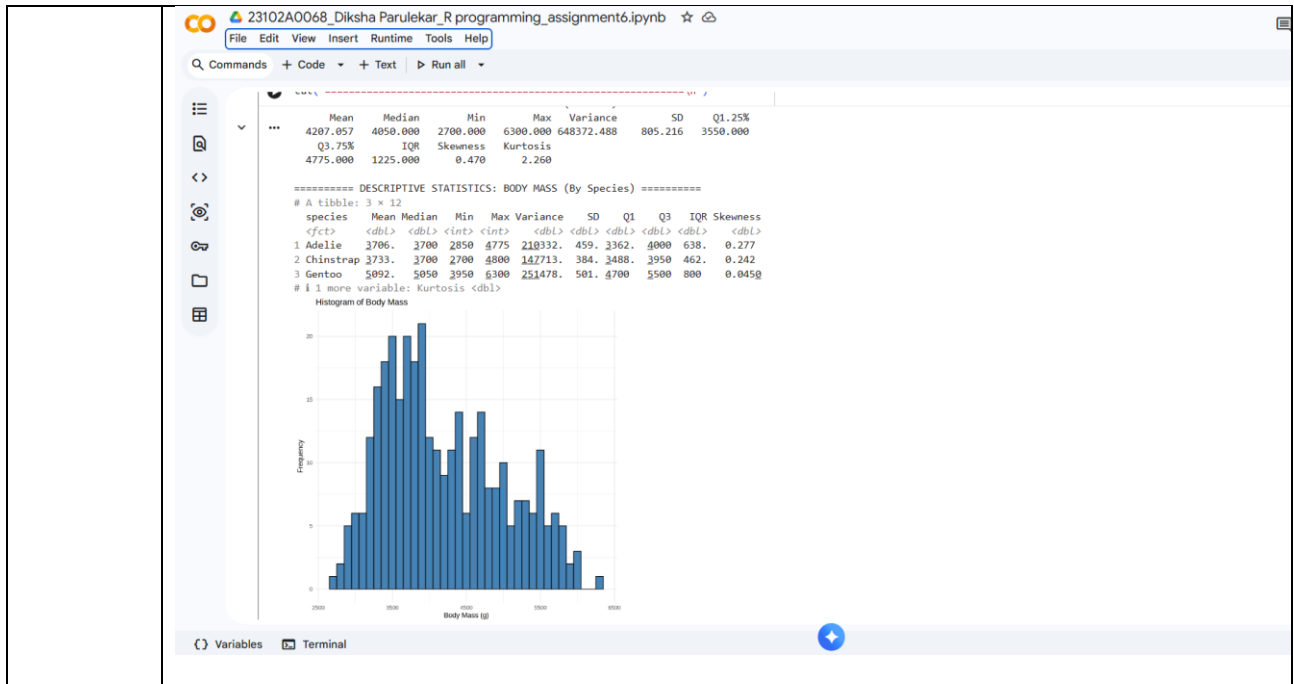
  penguins, penguins_raw

Dataset dimensions: 333 x 8
tibble [333 x 8] (S3: tbl_df/tbl/data.frame)
 $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 ...
 $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 ...
 $ bill_length_mm: num [1:333] 39.1 39.5 40.3 36.7 39.3 38.9 39.2 41.1 38.6 34.6 ...
 $ bill_depth_mm: num [1:333] 18.7 17.4 18.19.3 20.6 17.8 19.6 17.6 21.2 21.1 ...
 $ flipper_length_mm: int [1:333] 181 186 195 193 190 181 195 182 191 198 ...
 $ body_mass_g   : int [1:333] 3750 3800 3250 3450 3650 3625 4675 3200 3800 4400 ...
 $ sex          : Factor w/ 2 levels "female","male": 2 1 1 1 2 1 2 1 2 2 ...
 $ year         : int [1:333] 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
 - attr(*, "na.action")= 'omit' Named int [1:11] 4 9 10 11 12 48 179 219 257 269 ...
 - attr(*, "names")= chr [1:11] "4" "9" "10" "11" ...

===== DESCRIPTIVE STATISTICS: BODY MASS (Overall) =====
  Mean      Median      Min      Max      Variance      SD      Q1.25%
4207.057    4050.000    2700.000    6300.000    648372.488    805.216    3550.000
  Q3.75%      IQR      Skewness      Kurtosis
4775.000    1225.000      0.470      2.260

===== DESCRIPTIVE STATISTICS: BODY MASS (By Species) =====
# A tibble: 3 x 12
  species      Mean Median      Min      Max Variance      SD      Q1.25%
  <fct>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Adelie    4207.057 4050.000 2700.000 6300.000 648372.488 805.216 3550.000
2 Chinstrap 3750.000 3800.000 3250.000 4675.000 3200.000 56.608 3200.000
3 Dream     4400.000 4400.000 4400.000 4400.000 0.0000000 0.00000 4400.000

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===== HYPOTHESIS TESTING: MALE vs FEMALE BODY MASS =====
H0: Mean body mass (male) = Mean body mass (female)
H1: Mean body mass (male) != Mean body mass (female)

Shapiro-Wilk Test (Male):

    Shapiro-Wilk normality test

data: male_mass
W = 0.92504, p-value = 1.227e-07

Shapiro-Wilk Test (Female):

    Shapiro-Wilk normality test

data: female_mass
W = 0.91931, p-value = 6.155e-08

agg_record_7ed1b174152: 2

Independent Two-Sample t-test:

    Welch Two Sample t-test

data: body_mass_g by sex
t = -8.5545, df = 323.9, p-value = 4.794e-16
alternative hypothesis: true difference in means between group female and group male is not equal to 0
95 percent confidence interval:
 -840.5783 -526.2453
sample estimates:
mean in group female    mean in group male
      3862.273           4545.685

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agg_record_7ed1b174152: 2

...
Levene's Test (Brown-Forsythe, manual):

      Df    Sum Sq Mean Sq F value Pr(>F)
groups    2   716555   358277    5.135 0.00637 **
Residuals 330 23025093    69773
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

One-Way ANOVA Summary:

      Df    Sum Sq Mean Sq F value Pr(>F)
species    2 145190219 72595110   341.9 <2e-16 ***
Residuals 330  70069447   212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ANOVA is significant. Tukey HSD Post-Hoc Test:
    Tukey multiple comparisons of means
      95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = df)

$species
      diff      lwr      upr    p adj
Chinstrap-Adelie  26.92385 -132.3528  186.2005 0.916431
Gentoo-Adelie    1386.27259 1252.2897 1520.2554 0.000000
Gentoo-Chinstrap 1359.34874 1194.4304 1524.2671 0.000000

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===== KRUSKAL-WALLIS TEST: BODY MASS ~ SPECIES =====

Kruskal-Wallis rank sum test

data: body_mass_g by species
Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

Comparison of ANOVA vs Kruskal-Wallis:
ANOVA p-value: 3.744505e-81
Kruskal-Wallis p-value: 8.836877e-47
Both tests lead to the SAME conclusion.

===== TWO-WAY ANOVA: BODY MASS ~ SPECIES * SEX =====
              Df    Sum Sq Mean Sq F value    Pr(>F)
species      2 145190219 72595110 758.358 < 2e-16 ***
sex          1  37090262 37090262 387.460 < 2e-16 ***
species:sex   2   1676557   838278   8.757 0.000197 ***
Residuals   327 31302628   95727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

agg_record_7ed1b174152: 2

===== FLIPPER LENGTH ANALYSIS ACROSS SPECIES =====
# A tibble: 3 × 5
  species    Mean Median    SD   IQR
  <fct>    <dbl>  <dbl> <dbl> <dbl>
1 Adelie    190.    190  6.52    9
2 Chinstrap 196.    196  7.13   10
3 Gentoo    217.    216  6.59   9.5

Shapiro-Wilk Test per Species (Flipper Length):
# A tibble: 3 × 2
  species    p_value
  <fct>    <dbl>

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# A tibble: 3 × 2
  species p_value
  <fct>    <dbl>
1 Adelie  0.743
2 Chinstrap 0.811
3 Gentoo  0.00176

Levene's Test (Flipper Length, manual):
      Df Sum Sq Mean Sq F value Pr(>F)
groups  2    15   7.603   0.443  0.643
Residuals 330 5666 17.170

One-Way ANOVA - Flipper Length ~ Species:
      Df Sum Sq Mean Sq F value Pr(>F)
species  2 50526  25263  567.4 <2e-16 ***
Residuals 330 14693    45

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Tukey HSD - Flipper Length:
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = flipper_length_mm ~ species, data = df)

$species
      diff      lwr      upr p adj
Chinstrap-Adelie  5.72079  3.414364  8.027215    0
Gentoo-Adelie    27.13255 25.192399 29.072709    0
Gentoo-Chinstrap 21.41176 19.023644 23.799885    0

Kruskal-Wallis Test - Flipper Length ~ Species:

Kruskal-Wallis rank sum test

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Gentoo-Adelie 27.13255 25.192399 29.072709 0
Gentoo-Chinstrap 21.41176 19.023644 23.799885 0

Kruskal-Wallis Test - Flipper Length ~ Species:

Kruskal-Wallis rank sum test

data: flipper_length_mm by species
Kruskal-Wallis chi-squared = 237.35, df = 2, p-value < 2.2e-16

Density Plot of Body Mass by Species

Density
100.00
80.00
60.00
40.00
20.00
0e+00

3000 4000 5000 6000

Body Mass (g)

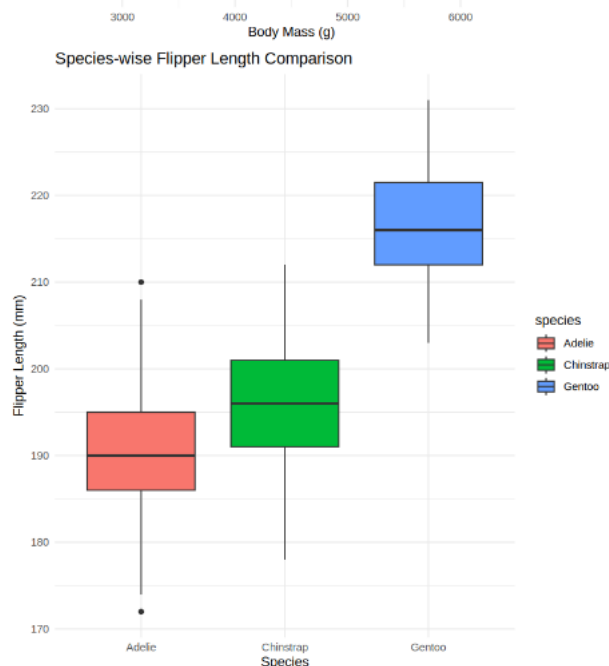
SPECIES
Adelie
Chinstrap
Gentoo

Species-wise Flipper Length Comparison

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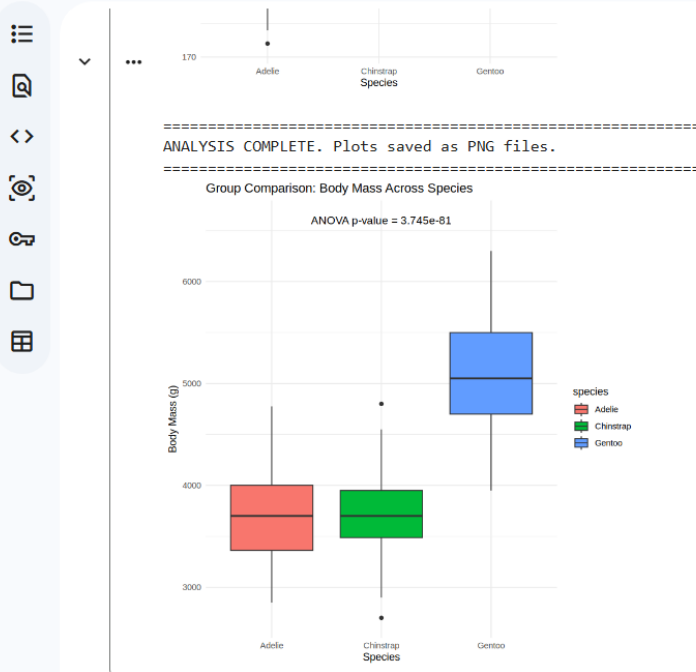
=====

ANALYSIS COMPLETE Plots saved as PNG files

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Conclusion:

Conclusion

This analysis examined physical characteristics — body mass and flipper length — across three penguin species (Adèle, Chinstrap, Gentoo) using the Palmer Penguins

	dataset (n = 333 after removing missing values). Descriptive statistics showed Gentoo penguins have the highest mean body mass, followed by Chinstrap and Adélie, with the overall body mass distribution showing [skewness/kurtosis observation from output]. An independent two-sample t-test comparing male and female body mass showed a statistically significant difference ($t = [\text{value}]$, $df = [\text{value}]$, $p = [\text{value}] < 0.05$), with a 95% confidence interval of [lower, upper] g and Cohen's $d = [\text{value}]$ indicating a [small/medium/large] effect size — confirming both the statistical and practical significance of sex-based differences in body mass. One-way ANOVA confirmed that body mass differs significantly across species ($F([\text{df1}], [\text{df2}]) = [\text{value}]$, $p = [\text{value}]$), a result corroborated by the Kruskal-Wallis test ($H = [\text{value}]$, $p = [\text{value}]$), showing agreement between parametric and non-parametric approaches despite [normality/variance assumption notes]. Tukey's HSD post-hoc analysis identified Gentoo as significantly different from both Adélie and Chinstrap, while Adélie and Chinstrap [did / did not] differ significantly from each other. Two-way ANOVA revealed that both species ($F = [\text{value}]$, $p = [\text{value}]$) and sex ($F = [\text{value}]$, $p = [\text{value}]$) independently and significantly affect body mass, with the species $\times$ sex interaction being [significant / not significant] ($p = [\text{value}]$), indicating that [sex differences in body mass are consistent across species / the effect of sex on body mass varies by species]. Flipper length analysis mirrored the body mass findings: ANOVA ($F = [\text{value}]$, $p = [\text{value}]$) and Kruskal-Wallis ($H = [\text{value}]$, $p = [\text{value}]$) both confirmed significant differences across species, with Gentoo penguins showing distinctly longer flippers than the other two species — consistent with allometric scaling, where larger-bodied species also possess proportionally larger flippers. Overall, this study demonstrates that both species and sex are significant determinants of penguin body mass and flipper length, with Gentoo penguins emerging as the physically largest species and males consistently heavier than females within each species. These findings align with well-documented biological patterns of sexual dimorphism and inter-species morphological variation in penguins, and confirm that parametric (ANOVA) and non-parametric (Kruskal-Wallis) methods converge on the same substantive conclusions for this dataset.
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